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# A deep learning-based novel model for masked face recognition

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**Abstract**

A major challenge in facial recognition is maintaining high accuracy without requiring individuals to remove their face masks. This research focuses on recognizing faces by analyzing the upper half—specifically the eyes, forehead, and visible side features like ears. We evaluated nine masked face recognition techniques, including MobileNet, MobileNetV2, ResNet50, DenseNet121, EfficientNetB0, InceptionV3, ResNet101V2, Xception, and our proposed model, using a dataset of ten classes. Our methodology introduces a novel approach that combines a modified MobileNetV2 with a convolutional architecture for fast feature embedding (CAFFE). This design was developed after extensive study of face recognition methods for artificial intelligence applications. To support this, we created a dataset of 3,340 masked face images using artificial masking and data augmentation techniques. The dataset was used to train and test our Python-based AI model. The core objective was to improve the model's performance by enhancing a pre-trained structure with 7 additional layers, optimized weights, and associated configuration files. Our proposed model integrates transfer learning, deep neural network (DNN) and deep learning (DL) architecture to extract image features of masked face classification (MFC) and masked face recognition (MFR) strategies to effectively extract features for masked face classification and recognition. In conclusion, our model outperformed the other eight traditional approaches, achieving 98.34% accuracy with a 1.66% error rate, while offering fast computation and reduced parameter usage. The source code used in our article are now made available on github platform https://github.com/AnilKumargithu/Masked-Face-Recognition-MFR-Classification-and-Prediction. Our created dataset of artificially masked face images utilized in this research is made available on Kaggle https://www.kaggle.com/datasets/banilkumar20phd7071/masked-face-recognition-dataset-artificial-masking.

**Keywords** Implemented AI · Application of AI · Deep neural network · Masked face classification · Masked face recognition · Transfer learning · Deep learning

## 1 Introduction

As masks hide much of the face and make it harder for observers to recognize a known face, it's likely that if more individuals of the anti-social elements begin wearing masks, there will be an increase in crimes and other criminal activities. The information obtained by using face recognition technology is useful for a number of other applications. Artificial intelligence (AI) is the simulation of human perception in computers, enabling machines to do tasks like learning, reasoning and problem-solving that usually require human intelligence. These consist of intelligent control, robot vision, human-computer interaction and AI (Shen et al. 2019). Most of the face recognition technologies that are now on the market are rather good. Individual factors, such as head position, skin colour and lighting conditions, still have an impact on the system's effectiveness (Deng 2021). Image-denoising approaches have the potential to enhance the overall efficiency of the system (Zeng et al. 2021). Along with hand geometry, iris scanning and fingerprint recognition, face recognition has become one of the most efficient techniques (Mou 2010; Anot and Singh 2016). Modern

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methods for extracting feature vectors from images of faces include the usage of DNN. The feature vector robustness can be impacted by several factors such as diverse locations, lighting conditions, partial occlusions and low-resolution noise (Tan and Triggs 2010). Through the use of a mouth-and-nosemask, the face recognition algorithm's useful information is reduced in half. In a study (Damer 2020), using all of the facial information has an impact on feature vector quality and decreases reidentification accuracy.

In the United States, the federal government organization known as the national institute of standards and technology (NIST) is responsible for developing and promoting measurement standards, especially those related to technology and cybersecurity for commercial facial recognition systems (Lane 2020). However, these algorithms decrease in accuracy when applied to the task of recognizing individuals using masks. Individuals lose the ability to be recognized by their unique facial features, such as their lips, nose and cheeks when they cover half of their face with a mask. The use of a facial recognition system that can detect concealed faces during security checks at airports, train stations and other restricted places will allow contactless checking, therefore reducing the transmission of infectious diseases. As a result, MFR research has become more popular due to its many benefits. Algorithms for recognizing faces use metrics including forehead-to-chin distance, facial curves and eye-to-chin distance to recognize individuals. A comparison is made between the features and the stored patterns, among the many people covered by this field of research are local, global and hybrid methods. Used local or geometrical feature-based approaches to assess size, shape and distance using crucial characteristics such as the nose, mouth and eyes (Napol´eon and Alfalou 2017; Bhattacharya et al. 2019).

In order to obtain features, primarily the most basic facial parts are segmented. The approach is not suitable for accurately representing facial emotions, occlusions, or placements (Liao et al. 2012). Convolutional neural networks (CNNs) utilize convolutional layers to gather input on spatial hierarchies. They are a kind of neural network used for pattern recognition and image processing with tasks involving computer vision. Face recognition using transfer learning of a paired CNN used to recognize a pair of two-face images sent into the Siamese network as input vectors (Heidari 2020). Labelled Faces in the Wild (LFW) is a widely used dataset for face verification and identification in computer vision, machine learning and deep learning applications. The network then applies a similarity criterion to the LFW dataset to determine whether or not both images belong to the same people (Heidari 2020). Siamese sparse representation classification (SiamSRC) is a face recognition system designed to improve matching and identification accuracy by combining Siamese networks with sparse information. The deep SiamSRC network uses block-sparsity face matching and has a small reference gallery consisting of domain-specific face images. Row sparsity is used to cluster unlabeled faces in videos from the target domain before deployment. Using pre-existing 3D face models that have cluster centers in the capture condition space, processing settings are used to create a limited set of synthetic faces for each source using representative intra-class data from the target domain. SiamSRC uses a block-structured sparse representation-based classification for pair-wise similarity matching utilizing facial images (Mokhayeri and Granger 2020).

A one-time usage A Siamese neural network (SNN) learning approach is offered as it only needs one training sample per class. Also, two instances of the same model with the same architecture and weights to train the network to identify similarities of vectors in photo images and is evaluated using the Euclidean distance (Sabri 2021). A SNN evaluates two parallel systems with distinct images by examining the absolute difference in the feature vectors to get the similarity score (Putra 2021). A smartphone facial recognition program made use of the Siamese CNN architecture can be used to compare two object descriptions. Siamese CNN is unique among one-shot learning techniques. An edge device with limited computational resources can benefit from MobileNet convolutional neural networks (CNN). The second version of MobileNet is MobileNetV2. ResNet is an acronym for Residual Network. RESNet50 is a deep convolutional neural network made up of 50 layers. The DenseNet is an acronym that stands for Densely Connected Convolutional Networks. A Deep Neural Network (DNN) that consists of 121 layers is known as DenseNet121. The baseline model is EfficientNetB0. Using a novel compound scaling method, it uniformly scales depth, width, and resolution to achieve high accuracy. The third version of Google's Inception architecture, also known as GoogleNet, is InceptionV3. The ResNet101V2 architecture has 101 layers and is a variant of the ResNet architecture. The term Xception refers to Extreme Inception (Aufar and Sitanggang 2022). The local binary pattern (LBP) is a texture descriptor in computer vision that is used to evaluate patterns in images. Applications such as texture classification and facial recognition frequently use LBP.

The SNN uses the uniform LBP technique and the frequency feature perception to detect faces in realworld settings. The LBP approach removes lighting effects in order to feed the network model vector-level data for eliminating target noise, the frequency feature perception efficiently compresses low-frequency features and swaps information with high-frequency features (Song and Ji 2022). Sparse representation classifiers (SRCs) are a kind of machine

learning and deep learning algorithm that characterize input samples using sparse linear combinations of basis vectors for face recognition and image classification. A variation of SRC called sparse representation classification with discriminative projection (SRC-DP) includes discriminative projection for improved results in tasks including face recognition and other pattern recognition tasks (Yang et al. 2013). For real-world face recognition promising results a SRCs with 10-fold cross-created not only offers a dimensionality reduction strategy that is suitable for SRC. SRC-DP improves performance by optimizing the ratio of the projected space's within-class reconstruction residual to its between-class reconstruction residual along face recognition by using low-dimensional human face representations (Yang et al. 2013). Proposed a novel anomaly detector that analyzes vehicle state over time by combining leading algorithms to identify associations in controller area network messages. The method, tested against state-of-the-art solutions, detects various attacks using only data from normal operating conditions (D 2023).

The following are the main contributions of this research:

i. Designed a modified MNV2 (MMNV2) architecture with 7 customized layers and developed a hybrid novel model by integrating CAFFE with the modified MobileNetV2 for effective photo image and real-time video predictions.
ii. A total 3,340 masked face images dataset of 10 classes is created by artificially masking, data augmentation techniques and made publicly available in Kaggle.
iii. Clearly addressed a gap in existing research related to masked face classification and recognition, with detailed processes for data collection and artificial masking.
iv. Evaluated the proposed model against 8 traditional models using accuracy, precision, recall, F1-score, error rate, and model parameters, with results shown in tables and confusion matrices.
v. Created a Google form in which 107 students participated, helping to test the model's performance in real-time videos under various lighting conditions.

The following is written in the later Sections of the present article: The relevant work has been described in Sect. 2. In this paper training, testing data visualization and distribution with data augmentation used is discussed in Sect. 3. Section 4, explains about modified MobileNetV2 model architecture implemented in our present research paper. Also describes the classification and recognition of masked faces by our proposed approach utilizing the CMNV2 model. In Sect. 5, the description is included of how well the evaluation performance of our CMNV2 approach outperformed the rest of the 8 different methods using a matrix of confusion, along with several various factors. Additionally, MFR classification and accuracy prediction results were discussed. Section 6 describes lastly, conclusion of paper is presented.

## 2 Related work

Two new algorithms using frontal images of humans achieved 90% success rate with near-greyscale template matching. The second technique used geometrical attributes like mouth position, nose width, length and chin shape (Brunelli and Poggio 1993). Developed an automated 3D face recognition technique that does not require occlusion. Images from Milano Bicocca University's 3D face databases and the Bosphorus were used in this study of partially hidden faces (Alyuz et al. 2013). To stop hackers from tricking facial recognition systems, the methodology used a 3D mask attack database (3DMAD) of realistic faces from the system database (Hamdan and Mokhtar 2018). Using the essential convolutional properties of occluded and non-occluded face categories, a pairwise differential siamese network (PDSN) is used to create a mask dictionary (Song 2019). Developed a cutting-edge deep learning and gradient facial recognition system for internet of things (IoT) security for occluded facial features (Mao et al. 2019). Designed a free software program named "MaskTheFace" that uses a database to rapidly mask faces and learn. Without the requirement for raw images, new features were added to pre-existing facial datasets to detect masked faces with high accuracy and few false positives (Anwar 2020). The proposed unique Latent component detection (LCD) technique for MFR is built in an end-to-end manner is possible to recognize and extract discriminative characteristics from the previously unknown, mask-resistant face area using both actual and fabricated training data (Ding 2020). Created an identity-aware mask generative adversarial network (IAMGAN) with a center-based cross-domain ranking method and a domain-constrained ranking (DCR) function to resolve the significant intra-class variance between masked and unmasked faces (Geng 2020).

A two-module end-to-end de-occlusion extraction technology is used to transfer the masked face recognition model completion approach (Li 2020). Continuous face recognition vendor test (FRVT) with and without masked faces was developed by the NIST for an FRVT verification benchmark, the NIST-information technology laboratory (ITL) evaluated face recognition performance utilizing digital masks and pre-release COVID-19 techniques on a large image collection (Ngan 2020, 2021). To replace 2D face recognition systems, 3D ones were developed, in which

3D face recognition databases respond better to changes in facial expressions (Adjabi et al. 2020).

Developed a model using many pre-trained recognition models capable of identifying faces with and without masks and assigning IDs, before being extracted and used for face recognition, for identifying in a single image (Alasadi 2020). The study made use of a masked face detection dataset (MFDD) with a simulated masked face recognition dataset (SMFRD) containing the largest real-world masked faces to evaluate facial recognition and detection (Wang et al. 2023; Huang 2021). Developed the first MFR challenge in an effort to evaluate deep facial recognition techniques with masked faces, using an online testing infrastructure, the MFR challenge's WebFace260M and insight face trackers analyze facial recognition models (Deng 2021). Designed a deep learning system that can recognize people wearing masks with accuracy. This study's ResNet-50 architecture is good at differentiating faces from masks (Mandal 2021). Developed an asymmetric-arc-loss that combines arc face and circle-losses to use an objective masked face recognition method (Yu 2021). Simulated, real and periocular-masked face datasets were created. To reliably recognize masked faces, these frameworks assess adjustments to hidden faces and periocular images (Mart´ınez-D´ıaz et al. 2021; Jeevan et al. 2022).

Using large-margin cosine loss (MFCosface), an MFR system was developed with an algorithm built to generate training data with masked faces utilizing fundamental facial features by multi-task cascaded CNN (Deng et al. 2021). Developed a facial recognition framework that can be used as a layer over current methods from the ground up with no impact on performance on common tests for face verification efficacy, both masked and unmasked (Neto 2021). Designed a RetinaFace method from the masked face's eyes, forehead and eyebrows that recognizes masked faces using DL and LBP features, for a multi-task learning facial detector the fact is both extra-supervised and self-supervised (Vu et al. 2022). Designed a technique for security cameras face (SCF) evaluation images that are blurred and are cleaned-real world masked face recognition dataset (C-RMFRD) database of ideal loss function network for synthetic masked face recognition (Hsu et al. 2022). Created 128-d encodings using deep metric learning and FaceMaskNet-21, DL techniques for the purpose of recognizing faces in photo images, real-time video streams and stable files (Golwalkar and Mehendale 2022). Mask-based DL face recognition was developed and evaluated on a masked face database using six CNN models (Shatnawi 2022).

Achieved 99% average accuracy on multiple malware families, outperforming state-of-the-art methods under both centralized and decentralized unbalanced data conditions (D'Angelo et al. 2023). Designed a mobile-friendly masked face recognition application using a DCNN which is small, fast and effective and used for MobileFaceNet and SeesawFaceNet evaluation (Prayogo 2023). A method was created using a recurrent neural networks (RNNs) for feature extraction and a histogram of oriented gradients (HOG) characteristic for previously hidden masked face problem with the development of the histogram-based recurrent neural network (HRNN) and DL techniques (Chong et al. 2023). Developed DL approaches, a pre-trained CNN named MobileNetV2 and a unique CNN architecture with these two models, can be identified between correct, incorrect and unmasked faces (Abdulmunem et al. 2023). This study explores various video classification tasks using CNN-LSTM autoencoders (CNN-LSTM-AEs), specifically by leveraging the feature extraction strength of convolutional neural networks and the temporal modeling and classification capabilities of deep neural networks. The combined architecture effectively captures compact spatiotemporal features for improved video classification performance (D 2021).

## 3 Experiments

The original images dataset is created by "Rajesh Kumar" and is available in the Kaggle (Kumar 2020) containing people of 100 classes without mask and augmentation of original sample images displayed in Fig.1. In this research paper randomly selected 10 classes out of a total of 100 classes as discussed in the below Sects. 3.1 and 3.2.



**Fig. 1** Dataset without face mask original sample images

### 3.1 Data distribution of training

Training data in Fig. 2 (a) helps the learning process and improves ability of models to predict. A total of 80% of the 2,675 images have been established to train 10 classes, giving model access to substantial amount of data that improves its ability to identify minute correlations and patterns. The 10 classes get the training data images in the following order: 'Aamir Khan' has 293 images, 'Aishwarya Rai' has 237 images, 'Akshay Kumar' has 250 images, 'Bipasha Basu' has 245 images, 'Govinda' has 300 images, 'Juhi Chawla' has 245 images, 'Kajol' has 245 images, 'Salman Khan' has 277 images, 'ShahRukh Khan' has 314 images and 'Shruti Haasan' has 269 images.

### 3.2 Data distribution of testing

A 20% of 665 images that were used to test the model are included in the testing set, as shown in Fig. 2(b). This is an essential stage in testing a model's predictability and accuracy. Separate recent occurrences from the model's training set using testing data, an accurate estimation of the models expected performance in real-world scenarios is given. 'Aamir Khan' has 73 images, 'Aishwarya Rai' has 59 images, 'Akshay Kumar' has 62 images, 'Bipasha Basu' has 61 images, 'Govinda' has 74 images, 'Juhi Chawla' has 61 images, 'Kajol' has 61 images, 'Salman Khan' has 69 images, 'ShahRukh Khan' has 78 images and 'Shruti Haasan' has 67 images. The purpose of testing data is to ensure that the model's predictions are correct and accurate when applied to new, unknown data, as well as to detect

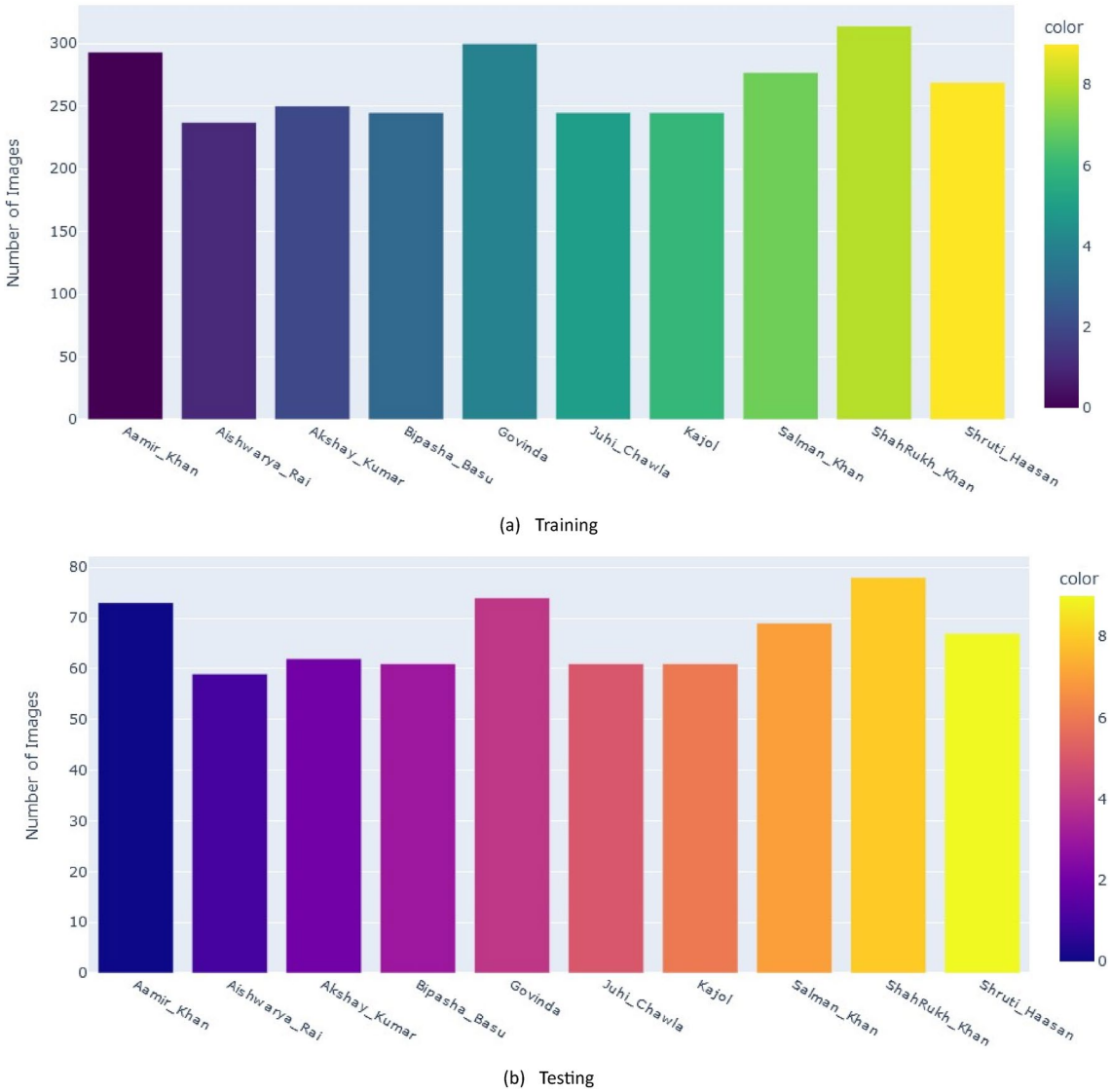



**Fig. 2** Data visualization

and evaluate any training errors, such as underfitting or overfitting.

### 3.3 Artificial masking and data augmentation

Artificial masking and augmentation techniques, such as horizontal flipping by $180^o$ and colour augmentation, are used in this suggested research. Through the use of augmentation techniques and masking people's faces, the original 1,297 images from 10 classes of non-masked faces are increased to 3,340 images, which are then used in our research for training and performance evaluation. This work's increased dataset is made available for the public free on Kaggle https://www.kaggle.com/datasets/banilkumar20phd7071/masked-face-recognition-dataset-artificial-masking.

## 4 Proposed methodology

The proposed method classifies and recognizes masked faces of images in photos and real-time videos using similarly proposed CMNV2 models (Kumar and Bansal 2023; Kumar and Misra 2024) and (Kumar 2025). Since the goal of our proposed CMNV2 solution is to perform MFC and MFR on images of photos and real-time video, respectively, a DNN architecture based on the TensorFlow, Keras and OpenCV libraries was developed. Transfer learning is the technique of utilizing a dataset to apply a model that was learned on one task to another that is comparable.

### 4.1 Modified MobileNetV2

Using the traditional MobileNetV2 architecture as a base, we built a customized MNV2 architecture in this research study (Howard 2017), adding 7 different layers while freezing the original MobileNetV2 all layers. For feature extraction, we used a transfer learning strategy. A detailed description is provided below, along with a technique is enhanced by adding personalized layers to the modified MobileNetV2 model as seen in Fig.3. The integration of updated MobileNetV2 for MFC and MFR is the next step in the CAFFE process (Jia 2014). A DL framework called CAFFE makes it possible to train and use DNNs quickly. It is popularly called for being quick, flexible, which makes this perfect to many different computer vision tasks. A pre-trained MobileNetV2 design from ImageNet (2021) and Keras applications (2020) provides the foundation for the improved MobileNetV2 architecture.

The architectural details of Fig. 3 is explained in depth in the following sections:

i. Input ($224 \times 224 \times 3$): A 224 by 224-pixel image having RGB (red, green and blue) colour channels.
ii. Frozen MobileNetV2 Layers: These layers serve as feature extractors and are an important part of the MobileNetV2 architecture. A DCNN called MobileNetV2 was developed for effective image classification. Crucially, the weights of these layers are "frozen," ensuring that they don't change during training. To make use of all of this appropriate data without the requirement for training, this research study uses a transfer learning technique.
iii. BatchNormalization: With normalizing the output of the previous layer, this layer improves training stability and convergence.
iv. GlobalAveragePooling2D: Added a GlobalAveragePooling layer after the BatchNormalization layer. The MobileNetV2 level feature map's spatial dimensions are reduced in part by this layer. It creates a fixed-size feature vector by averaging the data from each feature map. As it guarantees that features in the input image are consistently adapted regardless of size, this step is crucial.
v. Dense (256): Using a fixed-sized feature vector, this layer of 256 neurons is intended to learn complex patterns and features.
vi. Dense 1 (256): The recovered features are further refined by an extra Dense layer consisting of 256 neurons.
vii. Dense 2 (128): The process of refining features is carried out by a third Dense layer with 128 neurons.
viii. Dropout (0.5): A dropout layer with a rate of 0.5 is added after Dense 2. This dropout rate shows that during each training cycle, 50% of the neurons in this layer are randomly muted. This introduces randomness into the framework's learning process, which helps prevent overfitting.
ix. Dense 3 (10): The total number of neurons in the last, intended dense layer will ultimately match the total number of classes that the models output will be able to classify. The final output of the model is produced by this layer, which produces class outcomes for each of the 10 classes in the area of study.

### 4.2 Input data and augmentation

Trained and tested the proposed model using images from real-time videos and photos, as seen in block diagram of Fig. 4. The text files (TXT), opencv face detector.pbtxt, deploy.prototxt, res10 $300 \times 300$ ssd iter 140000.caffemodel, mask detector.model, opencv face detector.pbtxt each layer's weights maintained with different files is a part of suggested model.

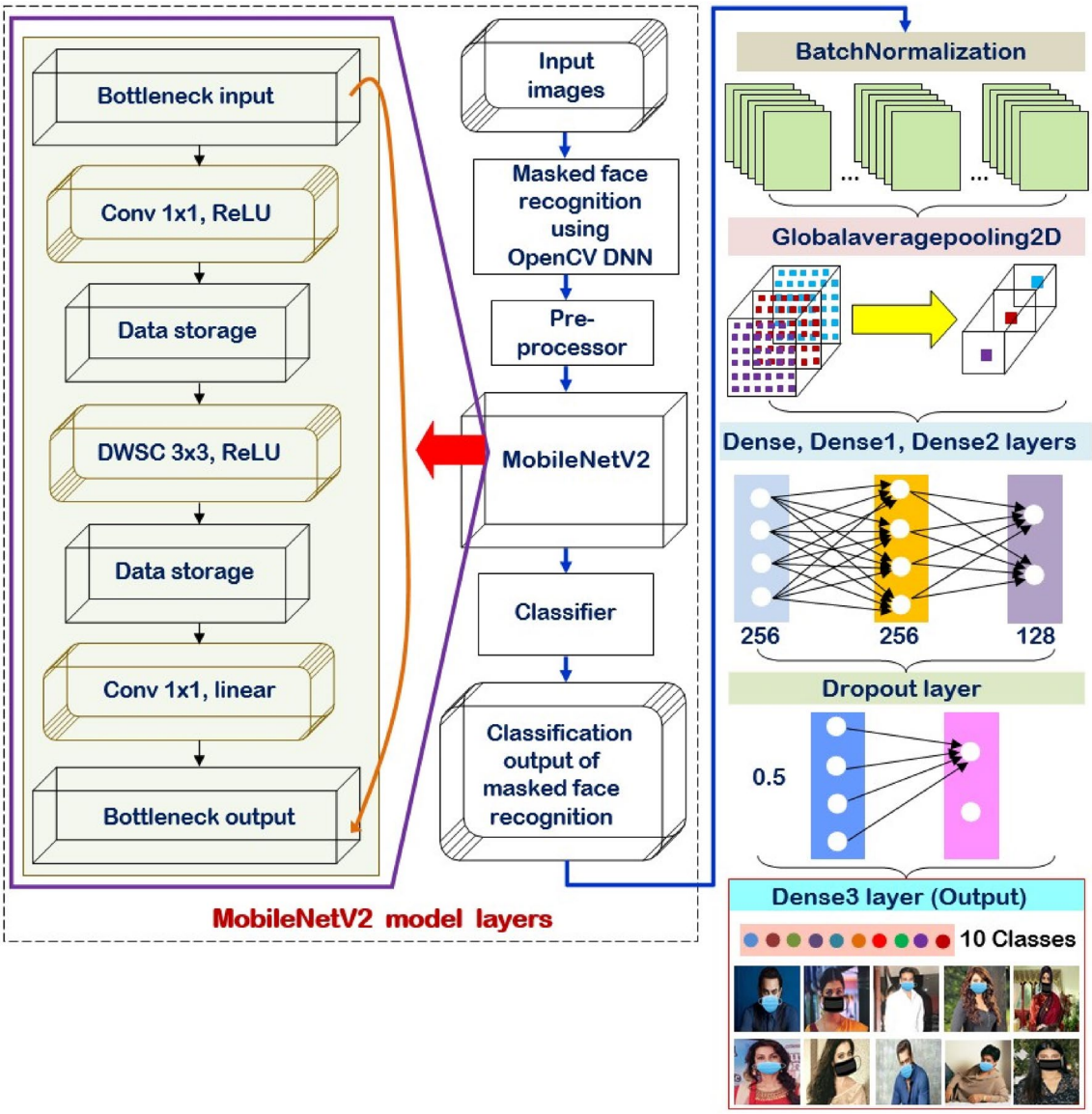



**Fig. 3** Modified MobileNetV2 architecture

The input process started with the collection of an initial dataset after downloading, consisting of images of unmasked people. This data collection builds the backbone of our research is described in Sect. 3. As a means of achieving the essential requirement for data with masked faces, technologically advanced data augmentation methods are used. Dataset of unmasked faces were artificially masked with the following blue and black colours two varieties of mask types and adjustments include a wide variety of viewpoints on emotions of face, conditions of lighting and colours background since thousands of images were created of masked faces as discussed in Sect. 3.3.

### 4.3 Data Pre-processing

As all faces have been properly masked enough. To make data retrieval easier in the next step, it was stored in a folder. A "pre-processing" application analyses every file, resizing the images accordingly, so that it can be used with the proposed approach in the designated folders. A graphical tool known as "labelling" was used to label the images once the data had been pre-processed. Further images are pre-processed with adding data after they imported and converted to array. Through procedure, values can be processed more quickly because images being changed to a Numerical Python (NumPy) library. After saving visual data, the label binarizer (LB) identified the numerical groups of 0 and 1 so that the pattern analysis could start.

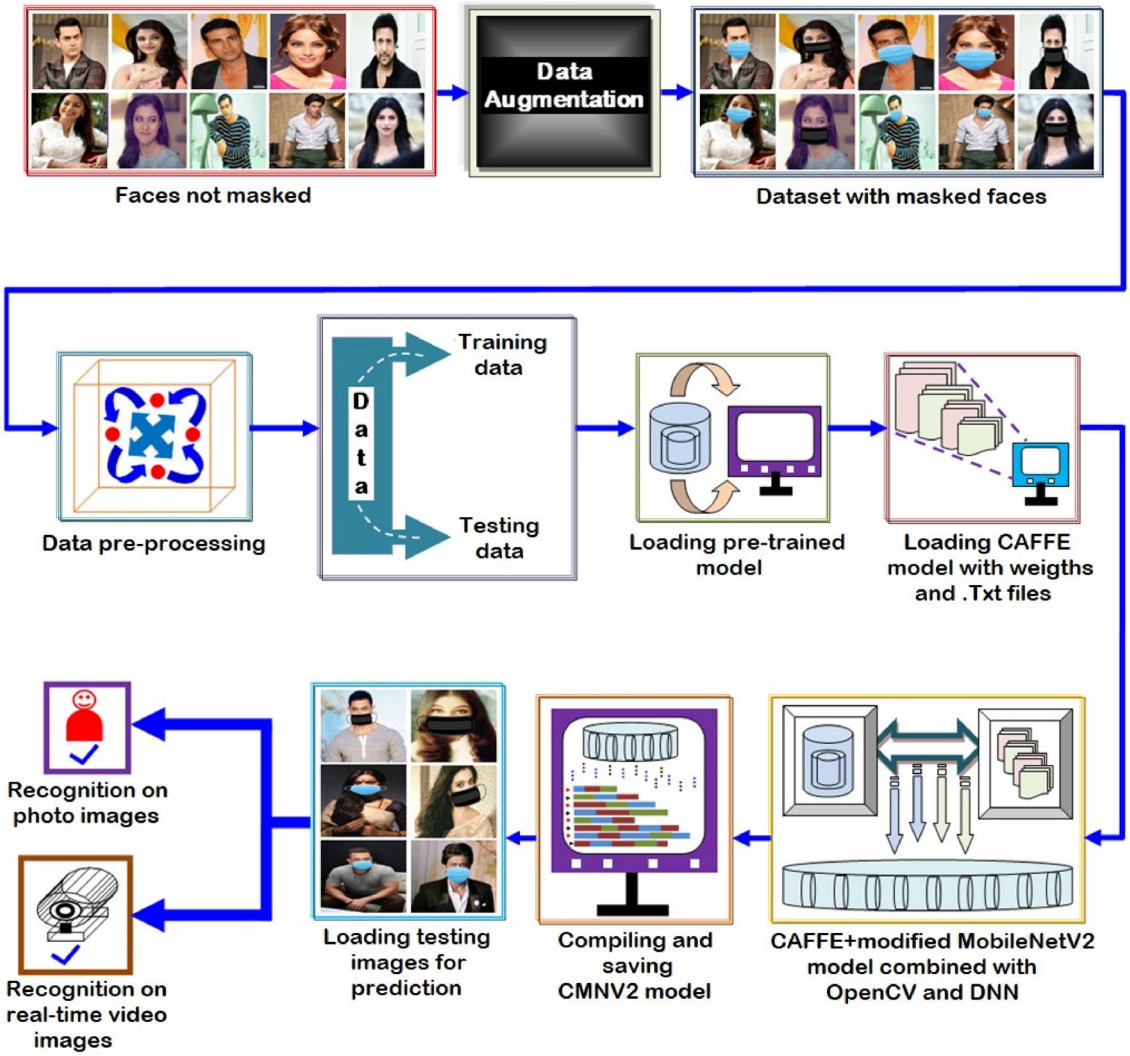



**Fig. 4** CMNV2 block diagram

### 4.4 Data splitting

The 3,340 artificially masked images of faces in the dataset are divided into two groups during data splitting: training and testing. As mentioned in Sect. 3, splitting the dataset into 80% for training and 20% for testing.

### 4.5 Model training

Proposed model methodology is described in Fig. 3. MobileNetV2 approach that had been trained earlier which imported from ImageNet and Keras applications. Transfer learning is used to create the 7 layers after loading the base model, as explained in Sect. 4. Next, by including an extra set of trainable CAFFE layers, features that can be used to classify and recognize a masked face were added. Weights were saved with two "TXT" and three "weight" files and the model was updated after the loading of the CAFFE. Making use of bias weights having minimal computational expense and not affecting earlier learned features is possible in the event utilize a pre-trained model. After that, we trained our suggested CMNV2 method for 60 epochs at a learning rate of 0.001 using 32 different training batches. The Adam optimizer modifies the speed at which each neural network weight advances towards a specified objective in deep learning and computer vision applications. The first and second gradient moments are estimated via the Adam optimizer. Rebalancing the weights of neural networks after each batch can be accomplished efficiently via transfer learning.

### 4.6 CMNV2 model

The CMNV2 model is suggested for masked face recognition and classification; it includes CAFFE with modified MobileNetV2. Using CAFFE framework for FaceNet

combined with modified MobileNetV2 framework for MaskNet, CMNV2 model applied for MFC and MFR after training. Blob is data structure that is used to pre-process and modify images in deep learning frameworks like CAFFE and OpenCV. Images are maintained in a multidimensional array format suitable for deep neural networks. Input data has to be normalized before being introduced into a neural network and this is done using a blob. OpenCV is a powerful computer vision toolbox with a number of real-time video and photo image analysis features. It analyses images and videos in combination with deep learning algorithms. Pre-trained face recognition and classification models from OpenCV are able to recognize faces in images and live video. Blob or areas of interest can be extracted from images using OpenCV, even ones with masked faces. OpenCV is used to pre-process images by resizing, normalizing and changing colour before applying them to a neural network.

### 4.7 Model testing

A 10 different classes of masked faces were given input to our proposed CMNV2 model, together with test data from both real-time video and photo images. Through calculating an object's bounding boxes of X and Y coordinates, the criteria are satisfied when the image detection probability crosses 0.5 and the masked face is detected in images such as photos and real-time video. To make sure that the bounding boxes are contained within the borders of the image, the class labels of the names are shown in red when the prediction is wrong and in green when the forecast is correct. These were used in the creation of the border and the text. Prototxt files, weights, blob function, RGB image dimensions, and real-time video files were combined to create MFR. There were also included bounding boxes and a label. Both the real-time video output and the predicted image results are shown in Figs. 9 and 11.

## 5 Results and discussion

Our model performed well in each test condition. Additional information about the face-specific problems with recognizing images for each class in the model is provided by the confusion matrices shown in Table 1.

Dataset contains 10 classes, as described in Sect. 4., of which 2,765 are training images (80% of total images) and 665 are testing images (20% of total images). The CMNV2 model was tested on 665 images and trained on 2,765 images. The accuracy rate of the CMNV2 model in recognizing masked faces is 98.34%. The confusion matrix is a useful technique in statistical analysis, ML and DL (Room 2019) for measuring the effectiveness of classification models. With 4 outcomes true positive ($T_{+ve}$), true negative ($T_{-ve}$), false positive ($F_{+ve}$) and false negative ($F_{-ve}$) the matrix shows the predicted and actual classes of a dataset. A positive class was predicted by the model with accuracy if $T_{+ve}$ is positive. The number of times the model correctly predicted a negative class is shown by $T_{-ve}$. The amount of incorrect predictions the model made is shown by the variable $F_{+ve}$. We investigated 10 models that generated an incorrect negative class prediction by using $F_{-ve}$. Confusion matrices of 9 distinct models evaluated as shown in Fig. 5 (a) to (i).

### 5.1 Accuracy

In overall classifying accuracy of the model is measured by it. This shows the percentage of occurrences based on the model categorized right. Equation (1), calculates classification accuracy.

$$\text{Accuracy} = \frac{T_{+ve} + T_{-ve}}{T_{+ve} + F_{+ve} + F_{-ve} + T_{-ve}} = 98.34\% \quad (1)$$

### 5.2 Error rate

It is defined as the percentage out of the total predictions that the model gives that are incorrect. This was given by the Eq. (2).

$$\text{Error Rate} = \frac{F_{+ve} + F_{-ve}}{T_{+ve} + F_{+ve} + F_{-ve} + T_{-ve}} = 1.66\% \quad (2)$$

### 5.3 Different models performance comparison

Table 1 provides testing data that was correctly predicted for 9 different models that were evaluated. Fig. 6(a) to (i) show the statistical analysis of precision, recall and f1-score of the 9 different models on testing data.

Our suggested CMNV2 model performs much better than 8 other models in design parameters and overall execution time. From Table 2 and Fig. 7(a) to (d), performance metrics of overall execution time of all 9 models, including design parameters of total, trained and non-trained are described with detail. With one of the lowest overall parameter counts in this sector, the CMNV2 approach has an excellent architectural design.

Among accuracy and error rates, the proposed model of CMNV2 outperformed 8 other models, as shown in Fig. 8. A crucial measure for a models predictive ability is accuracy, where CMNV2 achieved very high accuracy, suggesting that it is capable of producing accurate predictions.

**Table 1** Comparing evaluation performances of correctly predicted $T_{+ve}$ and $F_{+ve}$ testing values of 9 models

| Class Number | Class Name | MobileNet | | | MobileNetV2 | | | ResNet50 | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | Total | $T_{+ve}$ | $F+ve$ | Total | $T_{+ve}$ | $F+ve$ | Total | $T+ve$ | $F+ve$ |
| Class1 | Aamir Khan | 73 | 45 | 28 | 73 | 45 | 28 | 73 | 48 | 25 |
| Class2 | Aishwarya Rai | 59 | 50 | 9 | 59 | 49 | 10 | 59 | 54 | 5 |
| Class3 | Akshay Kumar | 62 | 50 | 12 | 62 | 58 | 4 | 62 | 59 | 3 |
| Class4 | Bipasha Basu | 61 | 58 | 3 | 61 | 57 | 4 | 61 | 58 | 3 |
| Class5 | Govinda | 74 | 56 | 18 | 74 | 70 | 4 | 74 | 72 | 2 |
| Class6 | Juhi Chawla | 61 | 53 | 8 | 61 | 56 | 5 | 61 | 56 | 5 |
| Class7 | Kajol | 61 | 46 | 15 | 61 | 47 | 14 | 61 | 50 | 11 |
| Class8 | Salman Khan | 69 | 56 | 13 | 69 | 55 | 14 | 69 | 65 | 4 |
| Class9 | ShahRukh Khan | 78 | 74 | 4 | 78 | 71 | 7 | 78 | 73 | 5 |
| Class10 | Shruti Haasan | 67 | 65 | 2 | 67 | 65 | 2 | 67 | 66 | 1 |
| **Class Number** | **Class Name** | **DenseNet121** | | | **EfficientNetB0** | | | **InceptionV3** | | |
| | | Total | $T_{+ve}$ | $F+ve$ | Total | $T_{+ve}$ | $F+ve$ | Total | $T+ve$ | $F+ve$ |
| Class1 | Aamir Khan | 73 | 59 | 14 | 73 | 56 | 17 | 73 | 65 | 8 |
| Class2 | Aishwarya Rai | 59 | 56 | 3 | 59 | 54 | 5 | 59 | 59 | 0 |
| Class3 | Akshay Kumar | 62 | 52 | 10 | 62 | 54 | 8 | 62 | 52 | 10 |
| Class4 | Bipasha Basu | 61 | 52 | 9 | 61 | 56 | 5 | 61 | 56 | 5 |
| Class5 | Govinda | 74 | 71 | 3 | 74 | 72 | 2 | 74 | 69 | 5 |
| Class6 | Juhi Chawla | 61 | 54 | 7 | 61 | 61 | 0 | 61 | 60 | 1 |
| Class7 | Kajol | 61 | 55 | 6 | 61 | 50 | 11 | 61 | 52 | 9 |
| Class8 | Salman Khan | 69 | 64 | 5 | 69 | 63 | 6 | 69 | 59 | 10 |
| Class9 | ShahRukh Khan | 78 | 77 | 1 | 78 | 75 | 3 | 78 | 74 | 4 |
| Class10 | Shruti Haasan | 67 | 65 | 2 | 67 | 65 | 2 | 67 | 64 | 3 |
| **Class Number** | **Class Name** | **ResNet101V2** | | | **Xception** | | | **CMNV2 (Proposed)** | | |
| | | Total | $T_{+ve}$ | $F+ve$ | Total | | $F+ve$ | Total | $T+ve$ | $F+ve$ |
| Class1 | Aamir Khan | 73 | 54 | 19 | 73 | 59 | 14 | 73 | 71 | 2 |
| Class2 | Aishwarya Rai | 59 | 57 | 2 | 59 | 57 | 2 | 59 | 58 | 1 |
| Class3 | Akshay Kumar | 62 | 59 | 3 | 62 | 61 | 1 | 62 | 62 | 0 |
| Class4 | Bipasha Basu | 61 | 54 | 7 | 61 | 52 | 9 | 61 | 59 | 2 |
| Class5 | Govinda | 74 | 70 | 4 | 74 | 71 | 3 | 74 | 72 | 2 |
| Class6 | Juhi Chawla | 61 | 60 | 1 | 61 | 57 | 4 | 61 | 61 | 0 |
| Class7 | Kajol | 61 | 54 | 7 | 61 | 56 | 5 | 61 | 60 | 1 |
| Class8 | Salman Khan | 69 | 64 | 5 | 69 | 67 | 2 | 69 | 67 | 2 |
| Class9 | ShahRukh Khan | 78 | 78 | 0 | 78 | 72 | 6 | 78 | 77 | 1 |
| Class10 | Shruti Haasan | 67 | 63 | 4 | 67 | 62 | 5 | 67 | 67 | 0 |

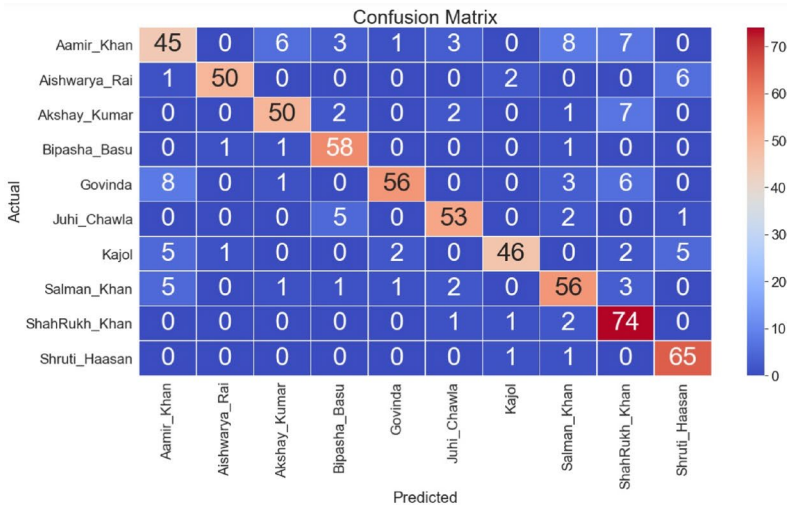
Compares multiple CNN models across 10 classes using Total, True Positives (T+ve), and False Positives (F+ve). MobileNet shows weaker performance with higher errors, while MobileNetV2 improves stability. ResNet50 and DenseNet121 provide strong and consistent accuracy with moderate errors, and InceptionV3 stands out with high T+ve and low F+ve. ResNet101V2 achieves high accuracy due to its depth but suffers from slightly higher misclassifications. Overall, the proposed CMNV2 model clearly outperforms all, showing the highest accuracy and the lowest errors across almost all classes

Output results of proposed model of CMNV2 provide a detailed analysis of the models ability to recognize masked faces in 10 different classes on photo images, as seen in Fig. 9. A bounding box especially green colour surrounding faces, with a green label displaying their respective names on top of each box, is used in Fig. 9(a) to (j) for accurately recognising masked faces. Exact location of identified face is shown in the green box. Simultaneously, it is clear from green label above bounding box that model recognized each face's mask effectively.

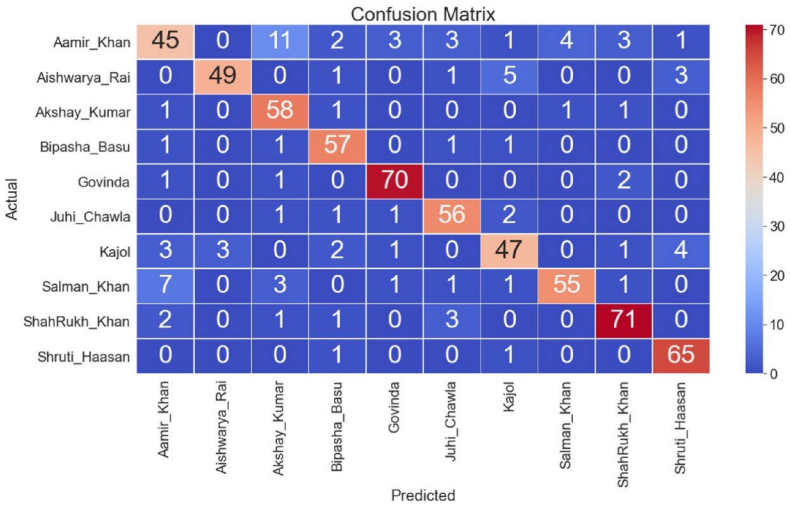
Using a specific red bounding box around the faces and a red label of their names displayed above each box, Fig. 9(k) to (p) incorrectly represents the recognition of masked faces. The recognized face is located inside the red box, which provides a visual reminder of its clear position. In addition, the red label placed on the bounding box indicates the models prediction, which is incorrect if it indicates that a mask is visible on each of the faces of different classes.

To address the need for real-world validation, we expanded our dataset by including actual masked face images. A total of 107 participants, including B.Tech students, voluntarily took part in testing the performance of our novel CMNV2 model for real-time video image processing and prediction is displayed in Fig. 10. To ensure participant comfort and privacy, masks were provided, and all testing sessions were conducted with proper precautions.

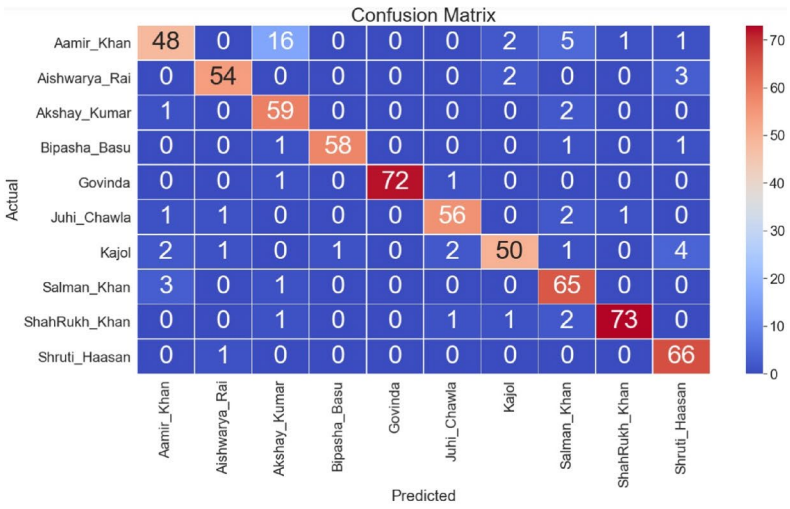
**Fig. 5** Confusion matrix of 9 different models



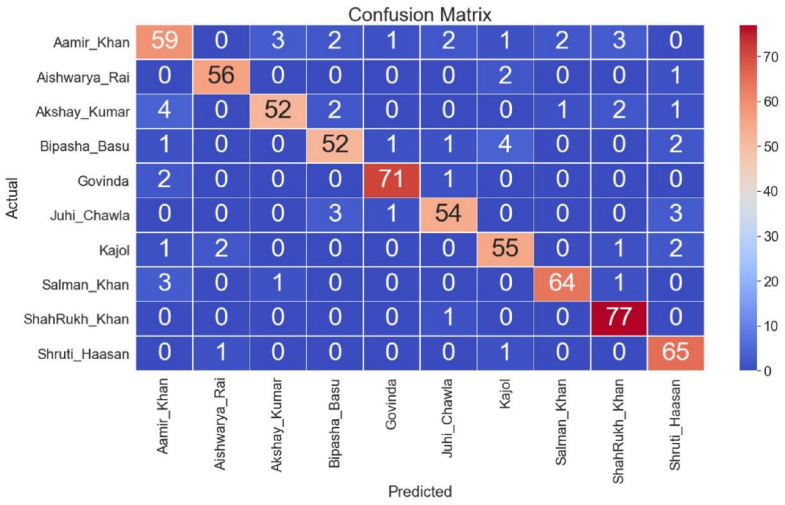

(a) MobileNet



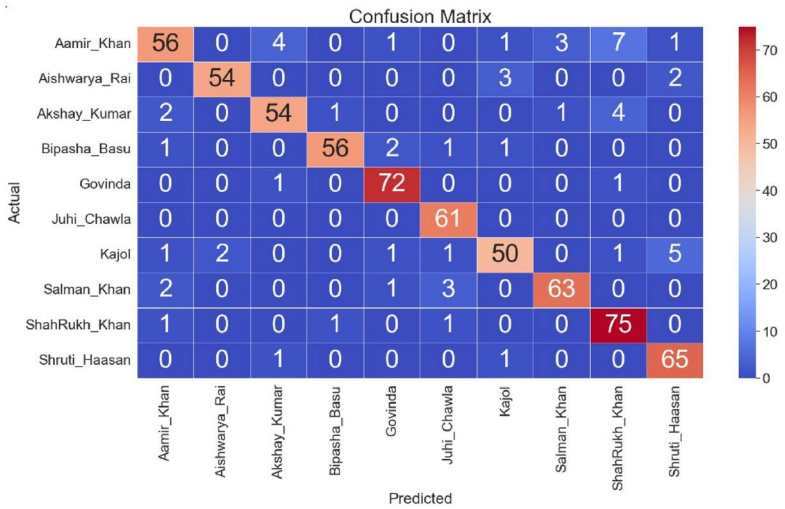

(b) MobileNetV2



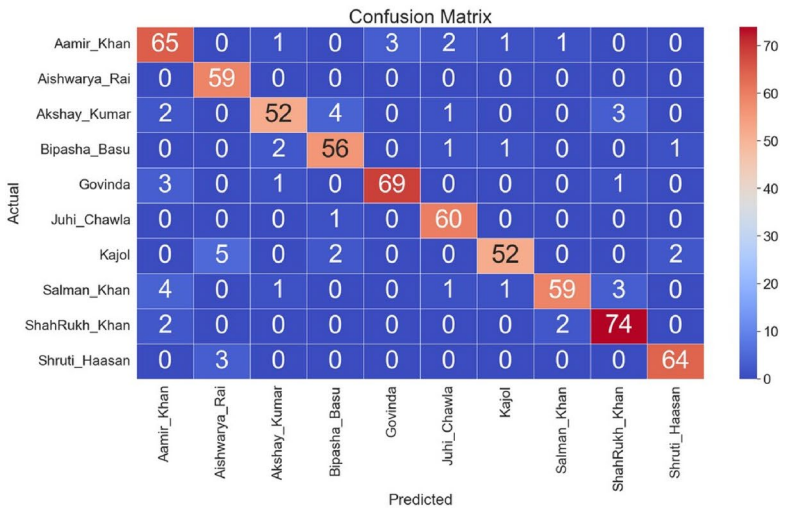

(c) ResNet50




(d) DenseNet121

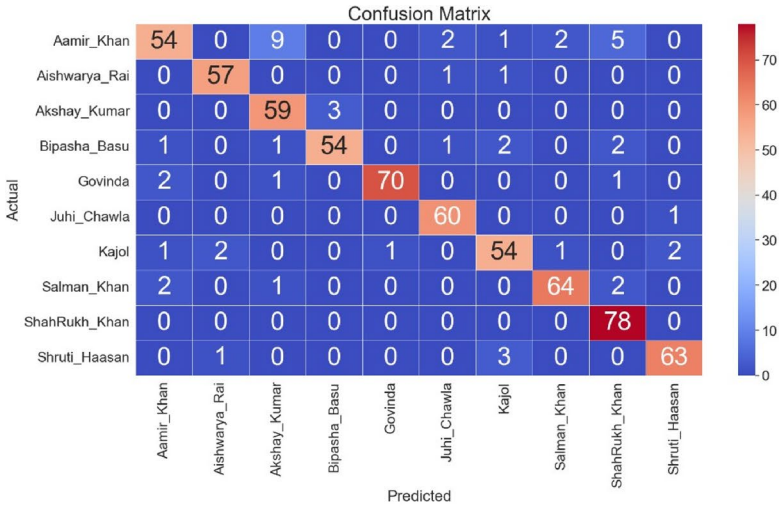



(e) EfficientNetB0

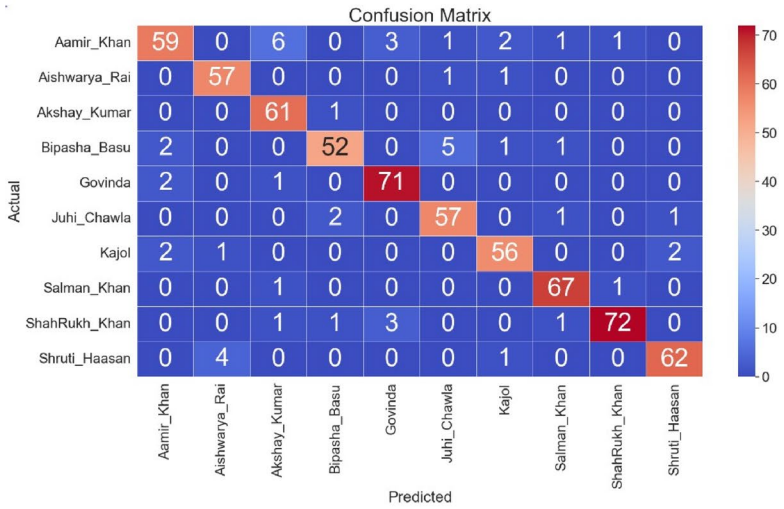



(f) InceptionV3

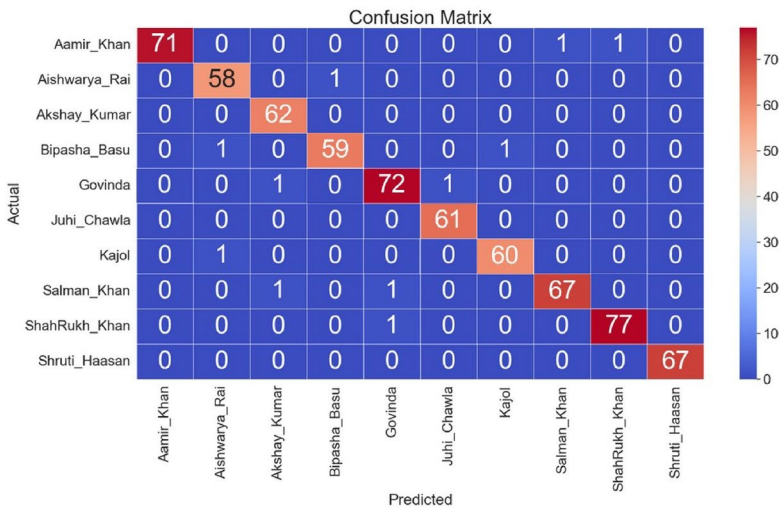
**Fig. 5** (continued)




(g) ResNet101V2




(h) Xception




(i) CMNV2 model (Proposed)

**Fig. 5** (continued)

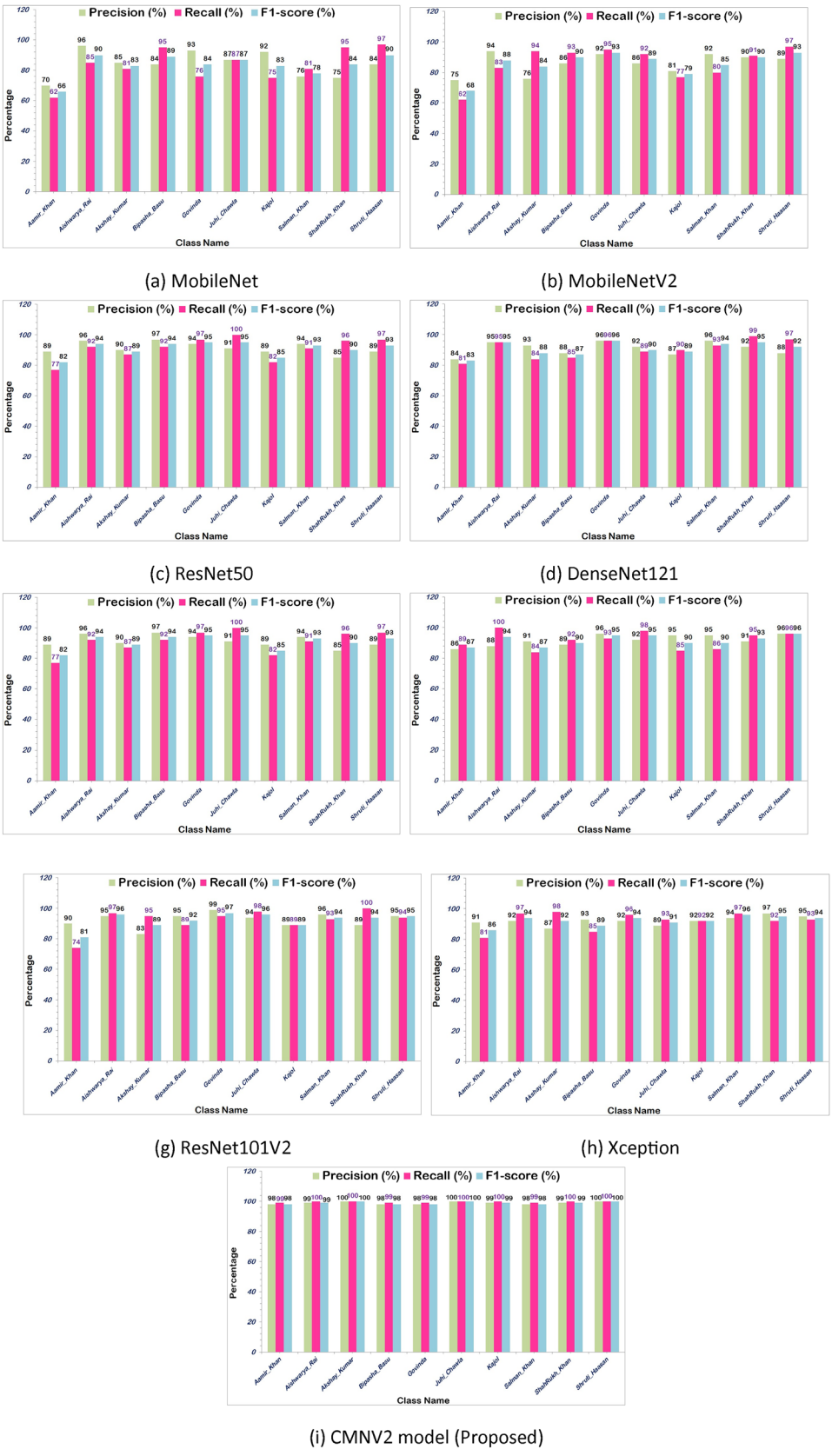



**Fig. 6** Statistical analysis of 9 different models

**Table 2** Comparing CMNV2 model parameters and runtime with 8 various methods

| Sl.No. | Model Name | Total Parameters | Trained Parameters | Non-Trained Parameters | Execution Overall Time (Seconds) |
|---|---|---|---|---|---|
| 1 | MobileNet | 32,39,114 | 32,17,226 | 21,888 | 24,960 |
| 2 | MobileNetV2 | 22,70,794 | 22,36,682 | 34,112 | 19,520 |
| 3 | ResNet50 | 2,36,08,202 | 2,35,55,082 | 53,120 | 93,750 |
| 4 | DenseNet121 | 70,47,754 | 69,64,106 | 83,648 | 41,280 |
| 5 | EfficientNetB0 | 40,62,381 | 40,20,358 | 42,023 | 30,570 |
| 6 | InceptionV3 | 2,18,23,274 | 2,17,88,842 | 34,432 | 89,520 |
| 7 | ResNet101V2 | 4,26,47,050 | 4,25,49,386 | 97,664 | 1,51,360 |
| 8 | Xception | 2,08,81,970 | 2,08,27,442 | 54,528 | 86,710 |
| **9** | **CMNV2 (Proposed)** | **26,94,858** | **4,36,874** | **22,57,984** | **8950** |

Compares CMNV2 with eight well-known CNN models in terms of parameters and execution time. Standard models like ResNet101V2 and DenseNet121 have very high total parameters and take longer execution times, making them computationally expensive. Lightweight models such as MobileNetV2 and EfficientNetB0 are faster but still have larger parameter counts compared to CMNV2. The proposed CMNV2 has the lowest trained parameters and one of the smallest total parameter counts, while also achieving the fastest execution time (8950 seconds). Overall, CMNV2 demonstrates superior efficiency with reduced complexity and faster runtime, making it the most resource-friendly model in the comparison

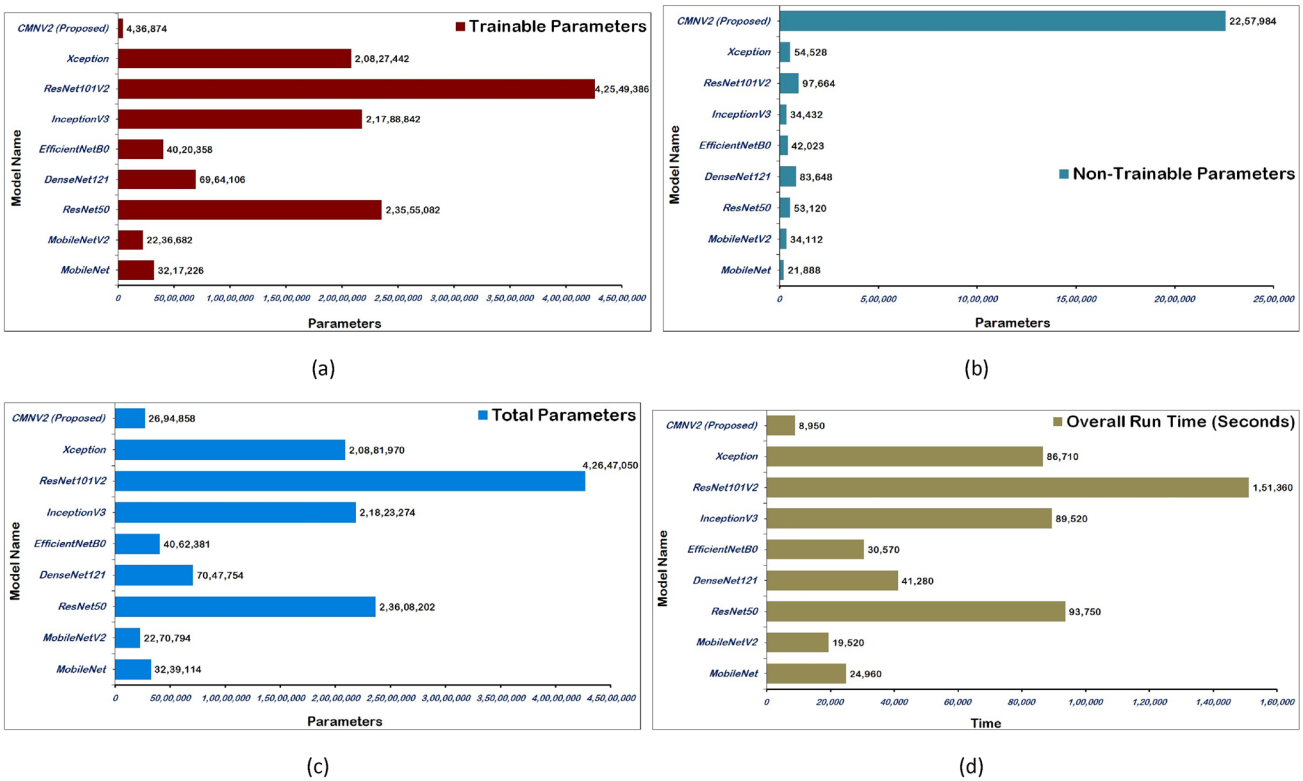
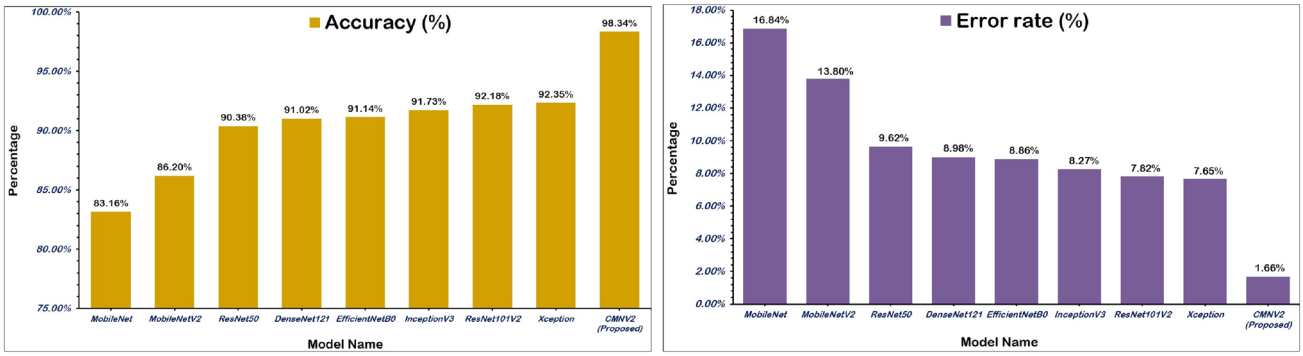



**Fig. 7** All 9 models performance comparison graphs

Each participant contributed by wearing a mask with partial occlusion, allowing us to evaluate the effectiveness of the CMNV2 model in real-time scenarios. The CMNV2 model was applied to real-time video frames to assess its prediction accuracy and reliability. The collaborative involvement of these participants played a crucial role in testing the model's functionality and robustness in realistic conditions, thus advancing research in the field of computer vision for masked face recognition.

In our proposed approach, each video frame is processed using the CMNV2 model at a frame rate of 30 frames per second (fps), which is widely supported by many systems, devices, and broadcasting standards. While higher frame rates like 60 fps require more power for streaming and storage, 30 fps offers a practical balance between smooth performance and visual quality. It ensures consistent video playback, reduces flickering, and results in smaller file sizes an advantage for mobile and storage-constrained platforms.




(a) (b)

**Fig. 8** Comparison graphs of accuracy and error rate for different 9 models

This frame rate also helps maintain visual clarity without requiring high-speed connections.

Figure 11 presents the real-time video results of the proposed CMNV2 model. It provides an in-depth investigation of the method's ability to recognize masked faces on real-time video images across 10 different classes.

The suggested CMNV2 models training and testing performance is shown in Fig. 12. Further, model learns to classify masked faces over time, which is illustrated in Fig. 12(a), which shows train accuracy by 'blue' colour and test accuracy by 'green' colour. An increasing training and testing accuracy curve shows the models recognition ability. Fig. 12(b) shows train and test loss. A reducing training loss shown with'blue' colour and testing loss with 'green' colour curve indicates that the model is improving at accurate predictions.

The CMNV2 model, which performs in multi-class classification, is used to predict 10 masked face recognition classes utilizing a receiver operating characteristic area under curve (ROC-AUC). The Fig. 13 shows model outperformed for recognizing among 10 classes, with a value greater than 0.5. The different rising curves for each of the 10 classes show how accurately the model classified the masked face classes. This curve determines the number of positive events also known as true positives, that a model can consistently detect, balancing sensitivity.

To enhance the model's feature extraction and classification capabilities for masked face age and gender identification, we incorporated seven additional layers into the modified MobileNetV2 architecture. These layers include Batch Normalization, Global Average Pooling 2D, two Dense layers with 256 units, one Dense layer with 128 units, a Dropout layer with a rate of 0.5, and a final Dense output layer for classifying ten categories. The selection of these layers was guided by their ability to introduce nonlinearity, prevent overfitting, and capture hidden patterns in the data. We conducted extensive testing with various layer configurations to evaluate the performance of the modified model. Our results consistently demonstrated that the inclusion of these specific layers significantly improved accuracy, accelerated convergence, and enhanced the model's ability to handle the complexity of facial features present in our dataset.

## 6 Conclusion

This paper presents the results of our proposed CMNV2 model for masked face classification and recognition. To support training, we developed a dataset of 3,340 masked face images using artificial masking and data augmentation techniques, which has been made publicly available on Kaggle. Our approach began by fine-tuning a deep neural network using the pre-trained MobileNetV2 architecture. We then enhanced it by adding seven additional layers through transfer learning to improve performance. The integration of the CAFFE architecture further accelerated training and improved model efficiency. We applied our dataset to train and evaluate nine models: MobileNet, MobileNetV2, ResNet50, DenseNet121, EfficientNetB0, InceptionV3, ResNet101V2, Xception, and our proposed CMNV2. The test accuracy of these models ranged from 83.16 to 98.34%. CMNV2 consistently achieved the highest accuracy (98.34%), lowest error rate (1.66%), fewer total and trainable parameters, and superior precision, recall, F1-score, and computation speed compared to the others. These results confirm that CMNV2 is the most effective solution for challenges in masked face classification and recognition. The model shows strong potential for real-world applications in areas such as education, healthcare, secure access, intelligent attendance systems, and various other industries.

Limitations: Our study observes a noticeable accuracy gap between still photo images and real-time video scenarios. While the model performs well on static images, it faces challenges in dynamic environments, particularly during

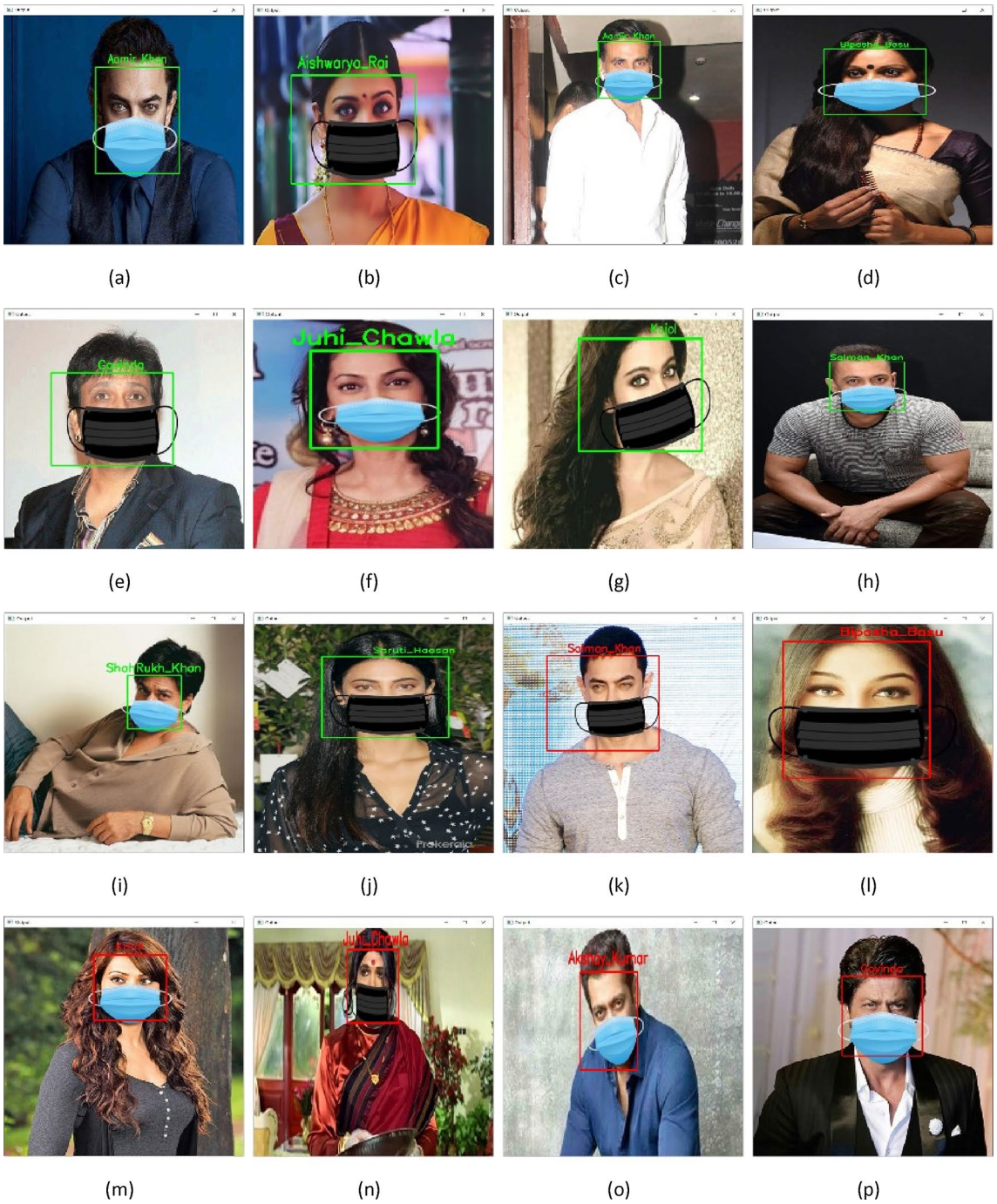



**Fig. 9** Masked face recognition of proposed CMNV2 model on photo images

real-time video processing. Factors such as sudden participant movements, geographical variations, and lighting changes impact the model's effectiveness. A specific limitation is the decrease in accuracy during abrupt participant motions, which introduces unpredictability and reduces the model's ability to maintain accurate predictions. This presents a significant concern for video-based applications, where rapid movements are common.

Future Work: Future efforts will focus on enhancing the model's performance in real-time video environments, especially under conditions of sudden movement. Potential improvements include incorporating motion compensation techniques and adapting the model architecture to handle dynamic input more effectively. Additionally, expanding the model's capability to estimate age and gender for multiple masked faces simultaneously in real-time videos will be

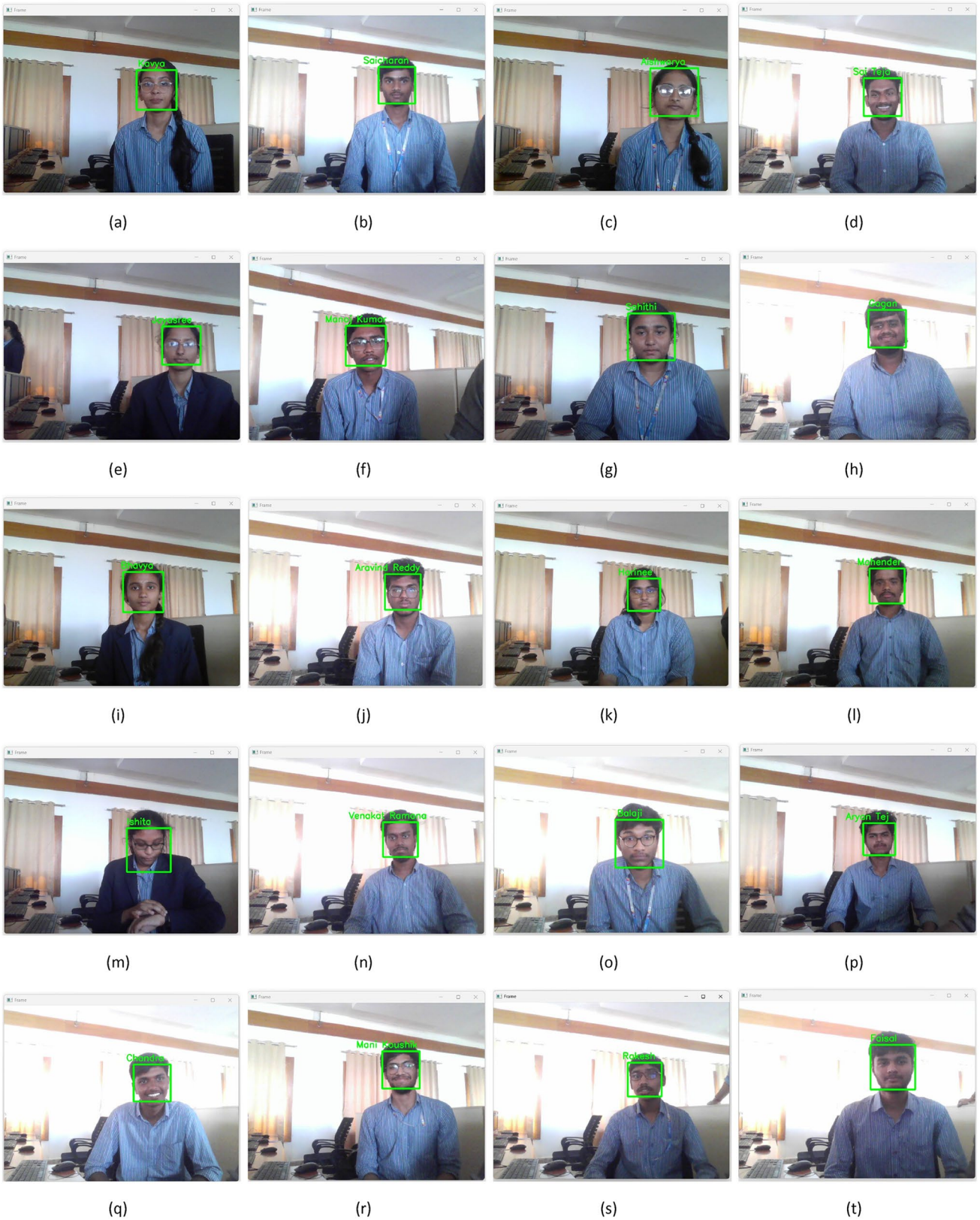



**Fig. 10** Proposed CMNV2 model face recognition on real-time video images

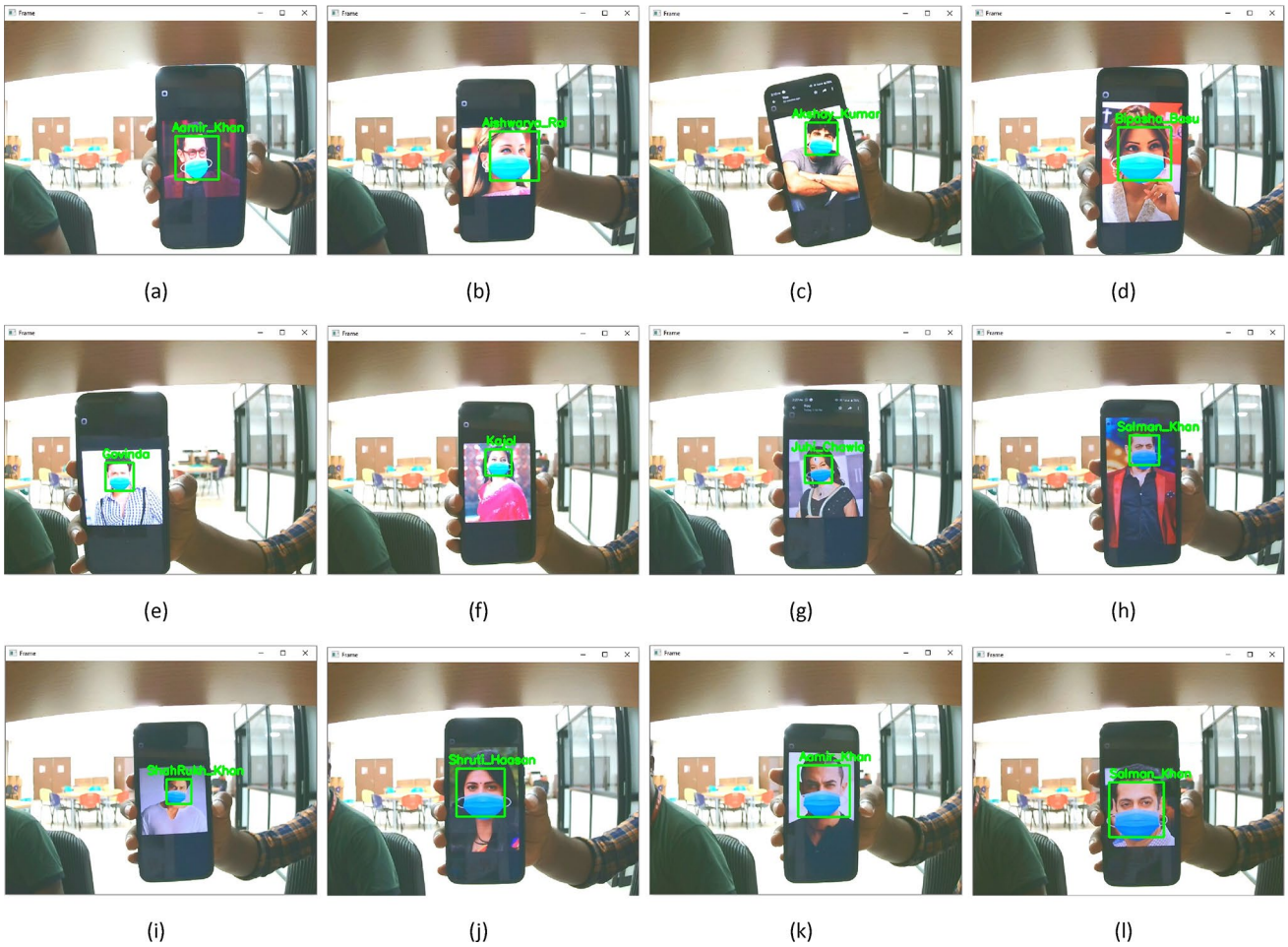



**Fig. 11** Proposed CMNV2 model masked face recognition on real-time video images using smartphone videos

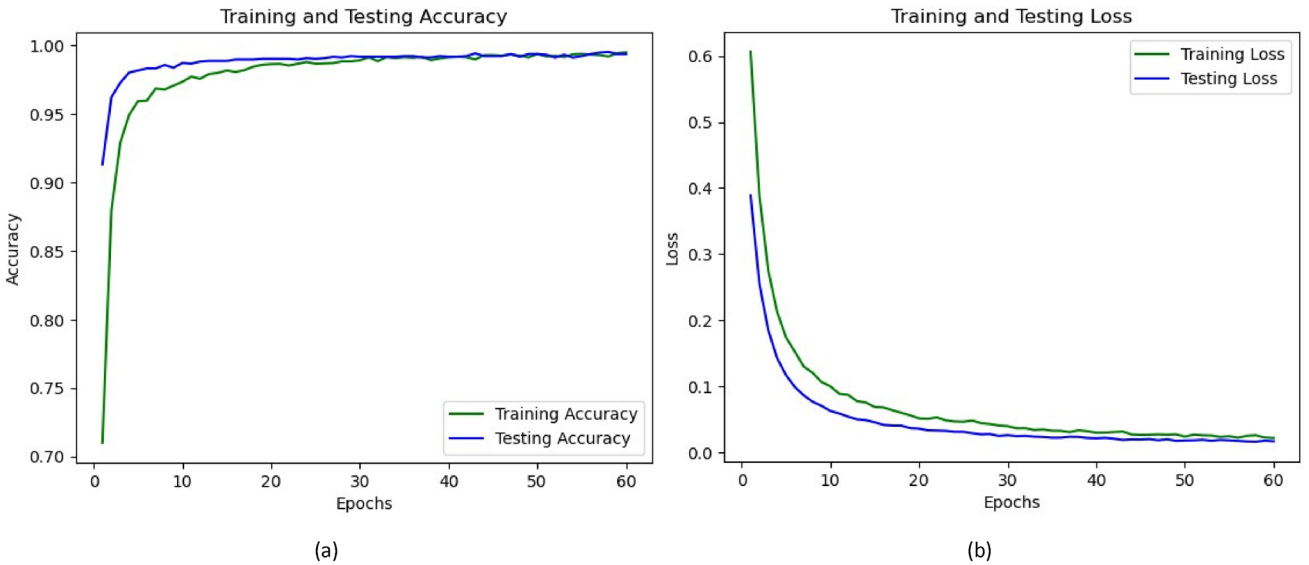



**Fig. 12** CMNV2 model accuracy and loss graph on training and testing

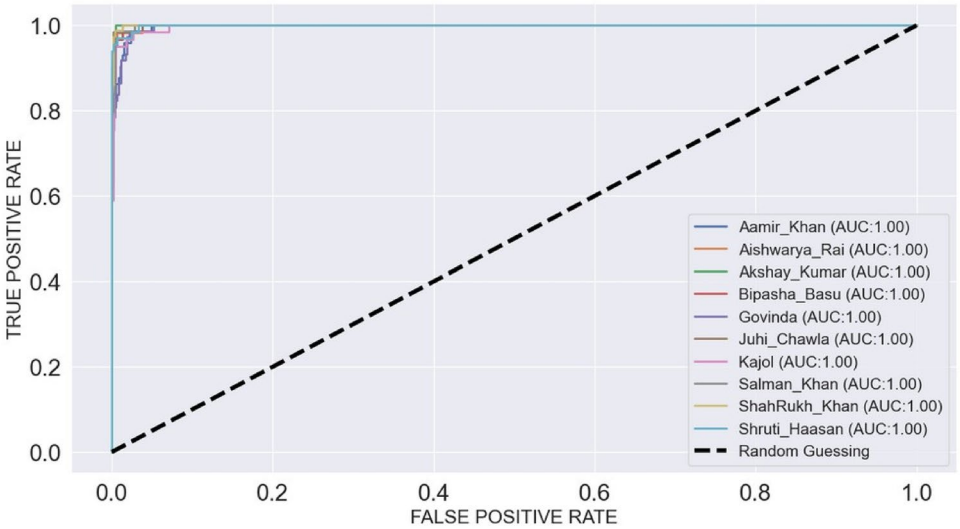



**Fig. 13** CMNV2 model ROC-AUC curve

explored. This advancement would be particularly valuable in settings such as crowded public areas and large group gatherings.

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**Author contributions** Dr. Neeraj Kumar Misra; Conceptualization, validation, formal analysis, research, resources, writing review and editing, visualization, supervision and project management. B.Anil Kumar; Conceptualization, methodology, software, writing, editing, original draught preparation. The final, published version of the paper has been reviewed and approved by all authors.

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**Data availability** This study's created artificially masked and augmented dataset, consisting of masked face images, is made available to the public on Kaggle https://www.kaggle.com/datasets/banilkumar20phd7071/ masked-face-recognition-dataset-artificial-masking. On request, the complete code will be provided.

## Declarations

**Conflict of interest** No authors have disclosed any conflicts of interest.

**Ethical approval** This article does not contain any studies with human participants or animals performed by any of the authors.

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